



Department of Computer Science and Engineering (Data Science)

B.Tech. Sem: III Subject: Statistics for Data Science

Experiment 4

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Date:	Experiment Title: Probability Distribution
Aim	To implement Probability Distribution using Python.
Software	Google Colab
Implementation	Using Python solve the following questions given below : 1. Out of 800 families with 4 pets(dog or cat) each, how many families would be expected to have (i) 2 dogs and 2 cats, (ii) at least 1 dog, (iii) at most 2 cats and (iv) both dog and cat. Code: <pre> from scipy.stats import binom k=800*(binom.pmf(2,4,0.5)) z1=800*(1-binom.pmf(4,4,0.5)) z2=800*(binom.pmf(0,4,0.5)+binom.pmf(1,4,0.5)+binom.pmf(2,4,0.5)) z3=800*1-800*(binom.pmf(4,4,0.5)+binom.pmf(4,4,0.5)) print(k) print(z1) print(z2) print(z3) </pre> <pre> 300.0 750.0 550.0 700.0 </pre> 2. The number of monthly breakdowns of a computer is a RV having Poisson distribution with mean equal to 1.8 . Find the probability that this computer will function for a month (i) without a breakdown, (ii) with only one break down and (iii) with atleast two breakdown.



Code:

```
{x} ✓ [7] from scipy.stats import poisson
0s poisson.pmf(0,1.8)

0.16529888822158653

✓ [8] poisson.pmf(1,1.8)
0s 0.2975379987988558

✓ 1-(poisson.pmf(0,1.8)+poisson.pmf(1,1.8))
0s 0.5371631129795577
```

+ Code + Text

3. A taxi cab company has 12 Ambassadors and 8 Fiats. If 5 of these taxi cabs are in workshop for repairs and an Ambassador is as likely to be in for repairs as a Fiat, what is the probability that

- (i) 3 of them are Ambassadors and 2 are Fiats,
- (ii) at least 3 of them are Ambassadors and
- (iii) all the 5 are of the same make.

Code:

```
✓ [0] from scipy.stats import hypergeom
0s hypergeom.pmf(3,20,5,12)

0.3973168214654289

✓ [27] 1-(hypergeom.pmf(0,20,5,12)+hypergeom.pmf(1,20,5,12)+hypergeom.pmf(2,20,5,12))
0s 0.7038183694530438

✓ [28] hypergeom.pmf(5,20,5,12)+hypergeom.pmf(5,20,5,8)
0s 0.05469556243550056
```

4. Buses arrive at a specified stop at 15 min intervals starting at 7 a.m., that is, they arrive at 7, 7:15, 7:30, 7:45 and so on. If a passenger arrives at the stop at a random time that is uniformly distributed between 7 a.m. and 7:30 a.m., find the probability that he waits

- (i) less than 5 min for a bus and
- (ii) at least 12 min for a bus.

Code:

✓
0s

```
from scipy.stats import uniform  
z1=uniform.cdf(5,0,15)  
print(z1)
```

0.3333333333333333

✓
0s

```
[30] z3=uniform.cdf(3,loc=0,scale=30) +uniform.cdf(18,loc=0,scale=30)-uniform.cdf(15,loc=0,scale=30)  
print(z3)
```

0.19999999999999996

5. The mileage which car owners get with a certain kind of radial tire is a RV having an exponential distribution with mean 40,000 km. Find the probabilities that one of these tires will last

(i) at least 20,000 km and

(ii) at most 30,000 km.

Code:

✓
0s

```
from scipy.stats import expon  
z1=1-expon.cdf(x=20000,scale=40000)  
print(z1)  
z2=expon.cdf(x=30000,scale=40000)  
print(z2)
```

0.6065306597126334

0.5276334472589853

6. In a certain city, the daily consumption of electric power in millions of kWh can be treated as a RV having an Erlang distribution with parameters $\lambda = 1/2$ and $k = 3$. If the power plant of this city has a daily capacity of 12 millions kWh, what is the probability that this power supply will be inadequate on any given day.

Code:

✓
0s

```
[32] from scipy.stats import gamma  
z1=gamma.sf(12,3,0,2)  
print(z1)
```

0.06196880441665898

7. Suppose, if in a basket there are balls which are defective with a Beta distribution of $\alpha = 2$ and $\beta = 5$. Compute the probability of defective balls in the basket from 20% to 30%.

Code:



	✓ 0s <pre>[33] from scipy.stats import beta z=beta.cdf(0.3,2,5)-beta.cdf(0.2,2,5) print(z)</pre> 0.23518500000000003 8. The marks obtained by a number of students in a certain subject are approximately normally distributed with mean 65 and standard deviation 5. If 3 students are selected at random from this group, what is the probability that at least 1 of them would have scored above 75 ? Code: ✓ 0s <pre>[34] from scipy.stats import norm from scipy.stats import binom norm.cdf(75,65,5) z3=binom.pmf(1,3,0.0227) + binom.pmf(2,3,0.0227)+binom.pmf(3,3,0.0227) print(z3)</pre> 0.06806490875099998
Conclusion	Probability distribution has been successfully implemented by using python library scipy.stats

Signature of Faculty